Form Events & Input Validation with DOM in JavaScript - Assignment Solutions

PW EARNERS : Coding and Web Development

By : Nishant Saini Sir

LEC - 41

Section 1 – Form and Input Events

1. Handle Form Submit Event

Question:
Create a simple form with a name input and submit button. Use the submit event to display a message when the form is submitted.
HTML:

```
<form id="myForm">
   <input type="text" id="name" placeholder="Enter your name">
   <button type="submit">Submit</button>
</form>
<p id="message"></p>
```

Solution:

```
let form = document.getElementById("myForm");
let message = document.getElementById("message");
form.addEventListener("submit", function(event) {
   event.preventDefault();
   message.textContent = "Form submitted successfully!";
});
```

Output:
Form submitted successfully!

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2. Prevent Form Submission

Question:
Create a form with a submit button and use preventDefault() inside the submit event to stop the page from refreshing.
HTML:

```
<form id="myForm">
   <input type="text" placeholder="Enter your name">
   <button type="submit">Submit</button>
</form>
```

Solution:

```
let form = document.getElementById("myForm");
form.addEventListener("submit", function(event) {
   event.preventDefault();
   console.log("Form submission stopped.");
});
```

Expected Result:
When the user clicks Submit, the page will not reload because preventDefault() prevents the default form submission behavior.

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3. Display Input Using the input Event

Question:
Create an input field and use the input event to display the value entered by the user in a paragraph.
HTML:

```
<input type="text" id="inputBox" placeholder="Enter something">
<p id="result"></p>
```

Solution:

```
let inputBox = document.getElementById("inputBox");
let result = document.getElementById("result");
inputBox.addEventListener("input", function(event) {
    result.textContent = `You entered: ${event.target.value}`;
});
```

Output:
If the user enters:
JavaScript
The output will be:
You entered: JavaScript

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4. Detect Changes Using the change Event

Question:
Create a <select> dropdown containing three programming languages. Use the change event to display the selected language.
HTML:

```
<select id="language">
    <option value="">Select Language</option>
    <option value="HTML">HTML</option>
    <option value="CSS">CSS</option>
    <option value="JavaScript">JavaScript</option>
</select>
<p id="selectedLanguage"></p>
```

Solution:

```
let language = document.getElementById("language");
let selectedLanguage = document.getElementById("selectedLanguage");
language.addEventListener("change", function(event) {
    selectedLanguage.textContent = `Selected Language: ${event.target.value}`;
});
```

Output when JavaScript is selected:

Selected Language: JavaScript

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5. Handle the focus Event

Question:

Create an input field and use the focus event to change its border or background color when the user clicks inside it.

HTML:

```
<input type="text" id="inputBox" placeholder="Enter your name">
```

Solution:

```
let inputBox = document.getElementById("inputBox");
inputBox.addEventListener("focus", function() {
    inputBox.style.border = "2px solid blue";
    inputBox.style.backgroundColor = "lightgray";
});
```

Output:

Before focus: Normal input style.

After focus: The input border and background color will change.

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6. Handle the blur Event

Question:

Create an input field and use the blur event to display a message when the user moves away from the input field.

HTML:

```
<input type="text" id="inputBox" placeholder="Enter your name">
<p id="message"></p>
```

Solution:

```
let inputBox = document.getElementById("inputBox");
let message = document.getElementById("message");
inputBox.addEventListener("blur", function() {
    message.textContent = "You left the input field.";
});
```

Output:

You left the input field.

Section 2 – Basic Form Validation

7. Validate a Required Name Field

Question:

Create a form with a name input. When the form is submitted, check whether the name field is empty. Display an error message if it is empty.

HTML:

```
<form id="myForm">
    <input type="text" id="name" placeholder="Enter your name">
    <button type="submit">Submit</button>
</form>
<p id="error"></p>
```

Solution:

```
let form = document.getElementById("myForm");
let nameInput = document.getElementById("name");
let error = document.getElementById("error");
form.addEventListener("submit", function(event) {
    event.preventDefault();
    if (nameInput.value.trim() === "") {
        error.textContent = "Name is required.";
    } else {
        error.textContent = "";
    }
});
```

Output when the field is empty:

Name is required.

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8. Validate Email Field

Question:

Create a form with an email input and check whether the email field is empty during form submission. Display an appropriate message if no email is entered.

HTML:

```
<form id="myForm">
    <input type="email" id="email" placeholder="Enter your email">
    <button type="submit">Submit</button>
</form>
<p id="error"></p>
```

Solution:

```
let form = document.getElementById("myForm");
let email = document.getElementById("email");
let error = document.getElementById("error");
form.addEventListener("submit", function(event) {
    event.preventDefault();
    if (email.value.trim() === "") {
        error.textContent = "Email is required.";
    } else {
        error.textContent = "";
    }
});
```

Output when the field is empty:

Email is required.

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9. Validate Password Length

Question:

Create a password field and validate that the password contains at least 6 characters.

HTML:

```
<form id="myForm">
    <input type="password" id="password" placeholder="Enter password">
    <button type="submit">Submit</button>
</form>
<p id="error"></p>
```

Solution:

```
let form = document.getElementById("myForm");
let password = document.getElementById("password");
let error = document.getElementById("error");
```

```
form.addEventListener("submit", function(event) {
    event.preventDefault();
    if (password.value.length < 6) {
        error.textContent = "Password must be at least 6 characters.";
    } else {
        error.textContent = "";
    }
});
```

Input:
Password: abc

Output:
Password must be at least 6 characters.

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10. Validate Multiple Form Fields

Question:
Create a form containing name, email, and password fields. On submission, check whether any of the fields are empty and display an appropriate error message.
HTML:

```
<form id="myForm">
    <input type="text" id="name" placeholder="Enter name">
    <input type="email" id="email" placeholder="Enter email">
    <input type="password" id="password" placeholder="Enter password">
    <button type="submit">Submit</button>
</form>
<p id="error"></p>
```

Solution:

```
let form = document.getElementById("myForm");
let nameInput = document.getElementById("name");
let emailInput = document.getElementById("email");
let passwordInput = document.getElementById("password");
let error = document.getElementById("error");
form.addEventListener("submit", function(event) {
    event.preventDefault();
    if (nameInput.value.trim() === "") {
        error.textContent = "Name is required.";
    }
    else if (emailInput.value.trim() === "") {
        error.textContent = "Email is required.";
```

```
    }
    else if (passwordInput.value.trim() === "") {
        error.textContent = "Password is required.";
    }
    else {
        error.textContent = "Form submitted successfully!";
    }
});
```

Example Input:
Name: Rahul
Email:
Password: 123456

Output:
Email is required.